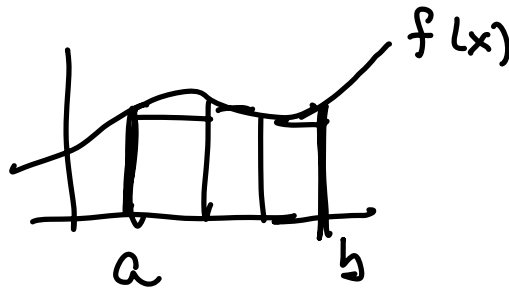


Class 36 The Riemann Integral

Goals

1. Know the basic idea of approximating a definite integral by a Riemann sum.
2. Understand terminology and notation for partitions, upper sums, and lower sums.
3. Know the definition of what it means for a function to be integrable on interval $[a, b]$.
4. Be able to use Darboux's Theorem to calculate a definite integral.

Basic Idea



Area of rectangle = _____ where

the base is _____ and

the height is _____

Dealing with bases

A partition of the interval $[a, b]$ is a finite set of numbers

$P = \{x_0, x_1, \dots, x_n\}$ where

If P and Q are two partitions of $[a, b]$, then Q is a refinement of P when

Example: Give three distinct partitions P , Q , and R of the interval $[1, 7]$ where Q is a refinement of P with at least one irrational number and R is not a refinement of P

$P =$

$Q =$

$R =$

Dealing with heights: A real number M is an upper bound for a set of real numbers S if and only if

If a set of real numbers S has an upper bound, then the supremum of S , denoted by ***sup* S** is the “least upper bound” satisfying two conditions:

1.

2.

Give examples of sets of real numbers satisfying

a) A set that is not bounded above.

b) An uncountable set, bounded above and containing its supremum of 4.

c) An uncountable set , bounded above but not containing its supremum of 5.

d) A countable set, bounded above but not containing its supremum of 6.

e) A countable set, bounded above and containing its supremum of 7.

Fill in similar definitions for lower bound and infimum for a set S .
Use lower case **m** to denote a lower bound for S and **inf S** to denote the infimum

Approximating integrals for a function f using partition $P = \{x_0, x_1, \dots, x_n\}$ for interval $[a, b]$.

Upper Riemann Sums

$$M_i(f) =$$

$$\Delta x_i =$$

$$U(f, P) =$$

Lower Riemann Sums

$$m_i(f) =$$

$$\Delta x_i =$$

$$L(f, P) =$$

$$L(f, P) \leq \int_a^b f(x) dx \leq U(f, P)$$

Example: Approximate $\int_1^7 x+1 \, dx$ using $P = \{1, 3, 7\}$ and $Q = \{1, 3, 5, 7\}$

P	upper height	width	Area of upper rect	$U(f, P)$
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P	lower height	width	Area of lower rect	$L(f, P)$
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Q	$M_i(f)$	Δx_i	A_i	$U(f, Q)$
$[1, 3]$				
$[3, 5]$				
$[5, 7]$				

Q	$m_i(f)$	Δx_i	A_i	$L(f, Q)$
$[1, 3]$				
$[3, 5]$				
$[5, 7]$				